

LAPORAN PROJECT BASED LEARNING

DEEP LEARNING

*Diajukan Sebagai salah satu syarat untuk menyelesaikan mata kuliah Deep Learning di
tahun ajaran 2025/2026*



Exercise Quality Assessment Using Autoencoder and Pose-Derived Statistical Features

Diusulkan Oleh:

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ABSTRACT

Accurate execution of exercise requires proper posture; however, professional coaching is often cost-prohibitive. This project proposes an automated assessment system utilizing computer vision and deep learning to objectively evaluate exercise quality and provide real-time feedback without wearable sensors. To accomplish this, the model used a media pipeline to improve robustness in human pose estimation, extracting 33 body key points from each video frame and handling variations in camera angle and body size by using a feature-extraction pipeline that computes pose-derived statistical features rather than raw coordinates. These features are used as input to the autoencoder, which is trained to reconstruct the pattern of the correct movement and posture using MSE. The system identifies the wrong form as an anomaly. This project developed four models for workout exercises, such as push-ups, squats, planks, and push-ups. Giving a clear separation between the correct and incorrect performance by a reconstruction error threshold that distinguishes the two classes. And this pipeline is deployed to a responsive web application built with React.js, giving users an experience for performing self-assessment using a standard web form and demonstrating a powerful derived statistical feature combined with an autoencoder that provides a robust, lightweight, and accessible solution for real-time work analysis.

Keyword: Deep Learning, Autoencoder, Computer Vision, Pose Estimation, MediaPipe, Web Application, Fitness Technology.

I. INTRODUCTION

Traditionally, human movements such as those in sports and exercise relied on professional expert observation. In sports and workouts, coaches and trainers analyze the movement of athletes carefully based on their knowledge and years of experience. They carefully analyze movement patterns to optimize technique, enhance performance, and reduce the risk of injury. These approaches are effective, but inherently subjective, dependent on individual expertise, and time-intensive.

The technological advancement in artificial intelligence, especially those in the healthcare and fitness sector provides an opportunity to transform how workout quality is evaluated. Advances in computer vision and deep learning allow objective, consistent, and repeatable assessments of workout movement, reducing reliance on human judgment alone. These advances open up opportunity for individuals to have assess their own workout activities without the need for time-intensive coaching.

Although a widely range of online fitness platform are available, many lack the ability to directly assess and score a person's workout. A guide on which workout to do and a guide on how to perform it, is simply not enough. Without accurate assessment, workouts might pose a significant risk of incorrect execution, which can reduce the effectiveness of training, slow rehabilitation progress, or even lead to injury.

This project aims to solve the problem by developing a platform and training a model for assessing workout quality using an autoencoder-based approach. Through this project, we aim to design a model capable of learning meaningful representations of human movement during exercise. The main idea is to use reconstruction error to distinguish between correct and incorrect movement patterns and evaluate how effective the approach is.

Through this project, we hope to significantly improve access to high-quality guidance in fitness and healthcare. This reduces the need of an expert, which might be inherently subjective and time-intensive. Through the project, we aim to support individuals to safely perform exercises at home while receiving constructive feedback and potentially improving adherence and recovery outcomes. With this work, we also hope to contribute to the growing field of unsupervised learning techniques with autoencoders, which is applied to a real-world movement analysis problems.

II. RELATED WORK

Previous research have shown Autoencoder ability in analysing squat position and giving visual correction. Using mediapipe to extract 3D landmarks from a camera, squat position can be classified into 7 different categories (1 correct and 6 specific errors). The study utilize Stacked Bidirectional GRU (Bi-GRU) with Attention Mechanism. The model achieved an accuracy of 94% and shows the effectiveness of Attention mechanisms and GRUs for handling the temporal data and evaluating exercise quality [1].

Other research shows LSTM Autoencoder architectures have the ability to detect fatigue and injury risk in athletes by identifying anomalies in movement. The model attempts to reconstruct the input with a high reconstruction error evaluated using mean square error (MSE) indicating an anomaly. The study use a dynamic threshold to flag the anomaly and shows that anomalies in healthy athletes appears at the end of sessions, while other patients that have injuries before, showed anomalies through the whole session. This research shows

that Reconstruction Error can be used as a statistical feature for quality assessment of movements without the need of negative data [2].

Liao et al. proposed a deep learning framework which integrates autoencoder based dimensionality reduction with statistical modeling for exercise assessment. Pose data dimensionality are reduced using a deep autoencoder approach and the model latent space are modeled using a Gaussian Mixture Model (GMM). A movement with a low probability or a low statistical likelihood score are considered as a low quality movement. The research shows the usage of statistical modelling is superior compared to the usage of simple thresholding [3].

Another research shows Autoencoders trained on correct movement patterns can be used to identify anomaly or deviation by analysing the reconstruction error. The research use 3D rotation angles of skeletal body parts as feature extraction mechanism rather than raw skeletal data. The study tested on Dense Low-Dimensional Autoencoder and Sparse High-Dimensional Autoencoder and calculate the reconstruction error using euclidean distance. The research shows the importance of using rotation angles of body parts as statistical features and the efficiency of Sparse Autoencoders to detect subtle kinematic anomalies in a high dimensional skeletal data [4].

Xiao et al. argues that feeding raw skeleton data directly into a model relies on implicit modelling which relies on the model to figure out the laws of physics and human motion from scratch. In the study Xiao et al. propose Explicit Modeling by manually calculating kinematic features that already follows the laws of physics before feeding the data into the model. The study use Walking Stride, Feet Displacement, and Neck Displacement feed into a probabilistic model. The study founds that using only 3 kinematic features, models managed to achieve similar or better results compared to models that used raw skeleton data [5].

The studies mentioned provide a strong foundation on statistical based feature extraction and the usage of autoencoders for movement quality assessment. The findings would become a foundation for our study in assessing the quality of workout exercise using Autoencoder and Pose-Derived statistical features

III. METHODOLOGY

The methodology for developing the “AI Trainer”, an AI-based system for evaluating exercise quality in push-up, pull-up, plank, and squat movements, is organized into two main

stages. The first stage focuses on feature extraction, while the second stage involves movement anomaly detection using a deep learning autoencoder model.

3.1 Dataset and Pre-processing

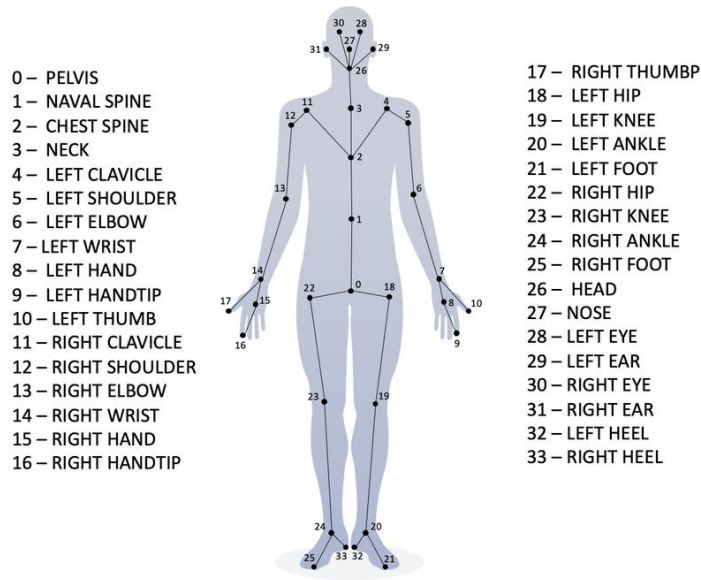


Image 1. Mediapipe Keypoint Distribution

The Process begins by utilizing MediaPipe Pose. This library is designed to detect 33 keypoints of the human body from each video frame, with a model complexity of 1. The videos used for the training process represent examples of movements that are considered correct. From the detected keypoints, the main features extracted are the joint angles relevant to the dynamics of each specific exercise. The angles are calculated using the reduced cosine formula:

$$\text{Angle} = \arccos\left(\frac{\overrightarrow{BA} \cdot \overrightarrow{BC}}{|\overrightarrow{BA}| \cdot |\overrightarrow{BC}|}\right) \quad (1)$$

It's important to note that while MediaPipe detects 33 points, not all points are used for every movement. The system identifies specific key points that most impact the quality of each exercise's form:

Exercise	Primary Keypoints Monitored	Focus
Push-up	Shoulders, Elbows, Wrists, Hips	Elbow extension and the alignment between the shoulder, hip, and knee.
Pull-up	Shoulders, Elbows, Wrists	Elbow flexion and shoulder movement to track the full range of motion.

Squat	Hips, Knees, Ankles	Knee flexion and hip depth to ensure proper squatting technique.
Plank	Shoulders, Hips, Knees, Ankles	Deviation from a linear alignment across the body to monitor core stability.

Table 1. Exercise Keypoints for Each Exercise

Each set of viewpoints generated throughout the video is then converted into five descriptive statistics, including mean, standard deviation, minimum, maximum, and PTP. This approach aims to capture both the spatial and temporal characteristics of the movement. This result in a 30 dimensional feature vector which contains the pose-derived stastical features.

3.2 Model Architecture

A feed-forward autoencoder is used for anomaly detection, with the goal of learning an efficient representation of normal movement patterns. This architecture includes an input layer, three encoder layers, a bottleneck (latent) layer, and three decoder layers for input reconstruction.

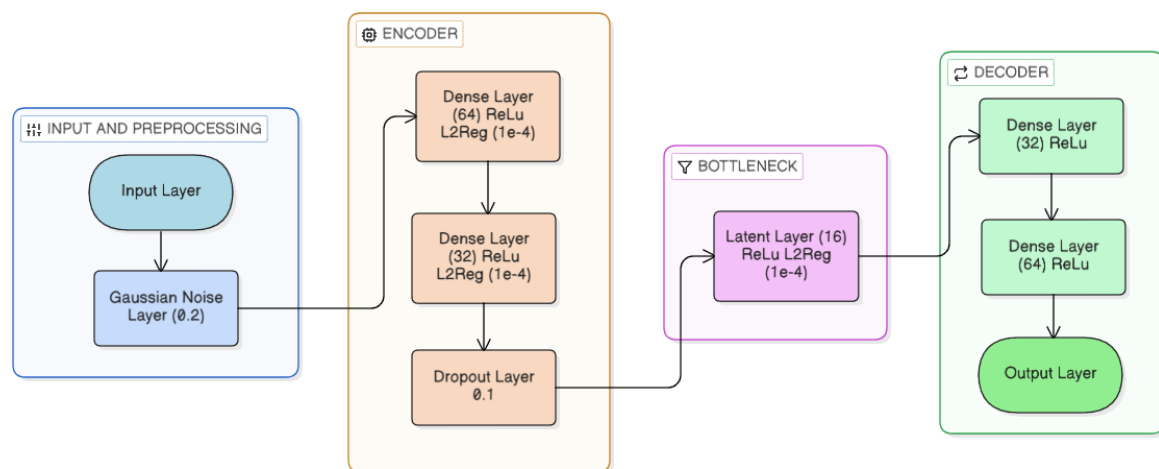


Image 2. Autoencoder Model Architecture

The autoencoder architecture used in this project includes:

1. The model starts with an input layer that receives normalized motion feature data, with the input dimensionality adjusting for the number of features used.
2. In the initial processing stage, a Gaussian Noise layer with a standard deviation of 0.02 is applied to increase the model's robustness to noise and reduce the risk of overfitting.

3. Next, the data is passed to a fully connected layer with 64 neurons and a ReLU activation function, equipped with an L2 regularization of $1e-4$ to reduce model complexity.
4. The next layer consists of a dense layer with 32 neurons and a ReLU activation function, which also uses an L2 regularization of $1e-4$ to improve generalization ability.
5. To prevent further overfitting, a dropout layer with a dropout rate of 0.1 is applied.
6. The feature representation is then compacted through a latent (bottleneck) layer consisting of 16 neurons with ReLU activation and an L2 regularization of $1e-4$, which serves to learn the core representation of the movement pattern.
7. In the decoder, the model uses a dense layer with 32 neurons and a ReLU activation function to begin reconstructing features from the latent space.
8. The reconstruction process continues with a dense layer containing 64 neurons and ReLU activation to restore the feature dimensions closer to their original form.
9. Finally, the output layer uses a linear activation function with the same number of neurons as the input, which aims to reconstruct the original motion features.

3.3 Training Configuration and Evaluation

The autoencoder model was trained using Mean Squared Error (MSE) as the loss function and the Adam optimizer with an initial learning rate of 10^{-3} . During training, the model learned to reconstruct its own input data using the same dataset as input and target (x_{scaled} , x_{scaled}). Key callbacks are implemented during the training process:

1. Early Stopping (with a patience of 12) prevents overfitting by stopping training when the validation loss plateaus.
2. ReduceLROnPlateau (with a patience of 6 and a factor of 0.5) automatically reduces the learning rate if validation loss improvement stalls.

The main metric for evaluation is the Reconstruction Error (MSE). Correct (normal) movements produce a low MSE, because the model reconstructs them accurately, while incorrect (anomalous) movements produce a high MSE.

3.4 Scoring Methodology

The scoring system utilizes four primary parameters derived from the training phase:

- MSE_{test} : The Mean Squared Error of the feature reconstruction for the tested video.

- P_5 : The 5th percentile of the MSE distribution from the "correct movement" training data, representing a high-performance baseline (L_{low}).
- P_{95} : The 95th percentile of the training data's MSE distribution.
- MSE_{max} : The maximum error threshold (L_{high}), which we defined as $P_{95} \times 3$ to establish a boundary for poor-quality movements.

The final score (S_{final}) is calculated using a piecewise linear function to map the MSE into a scale of 5 to 100:

$$S_{final} = \begin{cases} 100, & \text{if } MSE_{test} \leq P_5 \\ 5, & \text{if } MSE_{test} \geq MSE_{max} \\ 100 - 95 \cdot \frac{MSE_{test} - P_5}{MSE_{max} - P_5}, & \text{if } P_5 < MSE_{test} < MSE_{max} \end{cases} \quad (2)$$

We chose this specific approach to ensure the scoring is both fair and intuitive:

1. **Perfect Score (100)**: If a user's reconstruction error is lower than or equal to the 5th percentile of our training data, it indicates their form is nearly identical to the "correct" samples, and they receive a full score.
2. **Minimum Score (5)**: If the error exceeds the MSE_{max} threshold, the movement is considered an anomaly or highly incorrect, resulting in a minimum score of 5.
3. **Linear Mapping**: For any error falling between these two extremes, the score is calculated via linear interpolation. As the MSE_{test} increases, the score decreases proportionally from 100 toward 5. By using P_5 and MSE_{max} as benchmarks, the system adapts to the specific difficulty and variance of each exercise type.

IV. IMPLEMENTATION & RESULTS

4.1 Training Result

The training and validation loss curves for the Push-Up and Plank models exhibit remarkable convergence and stability. In both cases, the loss drops sharply in the first twenty epochs, with the training and validation lines nearly overlapping throughout. This tight alignment indicates a high degree of generalization, indicating that the models successfully capture the fundamental spatial and temporal features of these movements without overfitting. By maintaining nearly identical trajectories between the two datasets, the models are shown to

be able to filter out peripheral noise such as camera shake or minor anatomical variations while preserving the core movement signal.

In contrast, the Squat and Pull-Up models exhibit more complex learning behavior. The Squat training loss exhibits significant noise and fluctuation, indicating a higher level of variance or complexity in the movement data the model must navigate. Despite this volatility, the validation loss remains stable and slightly lower than the training loss, confirming that the model effectively identifies underlying patterns even amidst noisy input. Similarly, the Pull-Up model exhibits a small and consistent gap where the validation loss remains slightly above the training loss before reaching a plateau around the sixtieth epoch. While this suggests that the Pull Up data is more difficult to generalize than simpler training, the lack of divergence ensures the model remains robust.

The overall trajectory of the validation curves across the four models confirms the optimal balance between model capacity and generalization. Unlike many deep learning scenarios where validation loss eventually increases, a clear sign that the model is memorizing the training sample, these models exhibit validation curves that remain asymptotic to the training curve. The lack of upward inflection points confirms that the bottleneck layer of the autoencoder is appropriately sized: it is narrow enough to force meaningful feature compression yet large enough to retain important movement information. As a result, training terminates at a global minimum, ensuring the highest possible accuracy for previously unseen data.

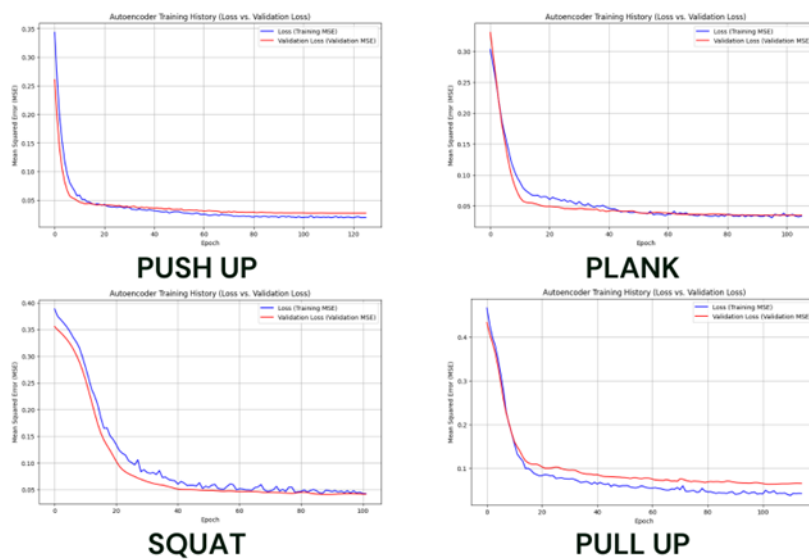


Image 3. Training and Validation Loss Across Workouts

4.2 Implementation

All the model trained are implemented in a simple web application developed using React.Js and Flask API backend. The application provides a user friendly interface, where users can record a video of their own or upload a video based on the workout chosen. The application will then send this information to the Flask API, where the video will go through the preprocessing pipeline mentioned above and then the score result will be returned and displayed back into the application. This simple web application act as an testing interface for users to test our model.

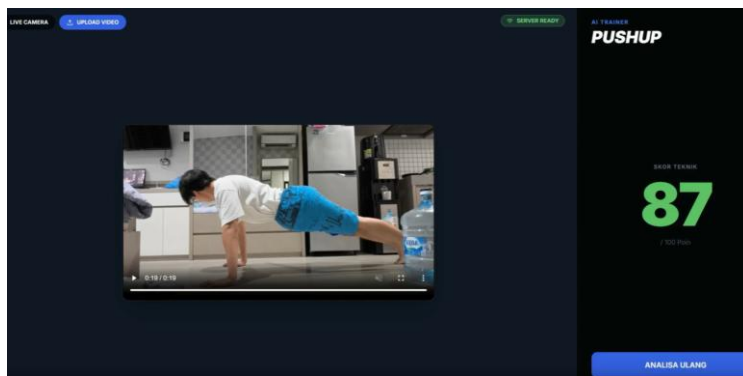


Image 4. Model Implementation on React.Js and Flask API

V. DISCUSSION & LIMITATIONS

5.1. Discussion

Our training results show that the autoencoder-based approach is quite effective in capturing the essence of the correct exercise form. By using pose-derived statistical features (such as mean joint angles and PTP) rather than raw skeletal coordinates, we successfully reduced input noise. This is likely why the Push-Up and Plank models converged so well; these exercises have relatively predictable, rhythmic, or static movement patterns that the model can easily "remember."

The higher variance observed in the Squat training history is an interesting discussion point. We believe this occurs because squats involve complex coordination between the hips, knees, and ankles, which creates more statistical variance in the feature set compared to static planks. Despite this, the model achieved a low MSE, indicating that it can still serve as a reliable basis for anomaly detection.

Similarly, the Pull Up model shows a slightly different trend, with a small but consistent gap between the training and validation losses. Unlike the Plank or Push Up, where the lines almost perfectly overlap, the Pull Up data appears to have higher complexity, likely due to the dynamic nature of the movement and potential body sway while hanging. However, because the loss stabilizes at a very low level around epoch 60, we are confident that the model is robust enough to provide accurate assessments.

The core idea of using reconstruction error as a "quality score" appears to be sound. Because the models are trained exclusively on "correct" movements, any significant deviations in form, such as a rounded back during a plank or a shallow squat, will theoretically result in a high MSE. This allows us to provide feedback without requiring a labeled dataset of "bad" movements, which is often difficult to collect.

5.2. Limitations

Despite the positive training results, our project has several limitations worth noting:

- **Sensitivity to Camera Placement:** MediaPipe Pose is quite robust, but its accuracy still depends on camera angle. If a user records their training from a frontal view while the model is trained on side-view data, the calculated angles and statistical features will change significantly, potentially leading to erroneous "false" scores.
- **Dataset Diversity:** Our training data is limited to a specific group of individuals. Factors such as different body proportions, clothing, or even background lighting can potentially affect keypoint detection and, consequently, reconstruction errors. A more "general" AI trainer would require a much more diverse dataset.
- **Subjectivity of "Perfect Form":** What we define as "correct" movement in our training set is based on our own understanding and the references we use. In reality, "correct form" can vary slightly depending on a person's mobility or fitness level. Our model currently uses a "one-size-fits-all" approach that may be too rigid for some users.
- **Real-Time Threshold Determination:** Although we achieved a low MSE (0.03 - 0.05), the biggest challenge remains in setting the appropriate threshold for what constitutes a "failure." Setting the threshold too low can lead to hesitation among users with small shape variations, while setting the threshold too high can miss true errors that could lead to injury.

VI. CONCLUSION & FUTURE WORK

6.1 Conclusion

The study have trained and implemented an autoencoder based architecture using pose derived satstical features to asses workout quality. The study sucsefully addressed the problem and limitations of traditional human coaching which are subjective and hindred in cost and availibility. Trough advances in pose detection frameworks, the study managed to extract pose keypoints from video and extract menaingful stastical features. The study have demonstrated that rich and meaningful stastical features can be extracted from raw pose keypoints.

The resutls have shown that the autoencoder architecture used in this study are capable of learning the patterns in workout for reconstruction. The model shows a low represenation error measured by MSE values that ranges from 0.03-0.05. This shows that during training, the model have sucsefully learn to reconstruct the stastical features from the video with very high accuracy. Training also shows that all four models have an optimal balance between model capacity and generalization as seen on the trajectory of the validation curves.

The trained model have also been sucseffuly impleneted with a simple web application using a combination of front-end and back-end framework of React.js and flask. The web application provides an accessible solution for the model to be tested using a video that can be recorded trough the application or using a pre-recorded video. In conclusion, the findings from this study have shown autoencoder and pose-derived features ability in assesing workout quality. We hope the project may contribute to the field of unsupervised learning and anomaly detection with autoencoder as a robust and reliable method to asses the quality of real-life workouts.

6.2 Future Work

Based on the findings and limitations of the study, this study reccomends the following improvents to be explored in future research :

1. Multiple view pose estimation, to overcome the problem camera placement and angle, future work should include a pose estimation technique that utilize multiple angles or other techniques that generate a view invariant features. Trough this the model trained would be more robust to changes in camera angles or position and imporve the model ability to minimize the reconstruction loss.

2. Dataset diversification, because the dataset used in this study are limited, a more diverse dataset with a wider demographic range, including different body types, ages, and mobility levels are recommended. The usage of other more advanced data augmentation techniques can also help improve model performance.
3. Error classification, an error classification mechanism would help users to see which part of the body during workouts are not being done correctly. A secondary classification model that can classify the error made by the user that contributed the most to the reconstruction error would provide better feedback to the user.
4. Adaptive thresholding, to address the issue that each user workout are different depending on their demographic, an adaptive thresholding mechanism can be used. Model can adjust thresholds based on the user's specific demographic or fitness level, hence further personalizing the workout assessment.

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APPENDIX

Team Contribution Statement:

Name	NIM	Contribution
Jonathan Carlo	2702210213	Video Editing, Report, Model Development
Bren Alden	2702242594	Model Development, Report
Andy Saputra	2702234094	Web Development, Report

Github Repository:

<https://github.com/Andysaputra1/deeplearning>

Video Demo:

https://drive.google.com/drive/folders/1MJklDV9-p-8SHSIUZAeH_W9UZ_h1N19I?usp=sharing